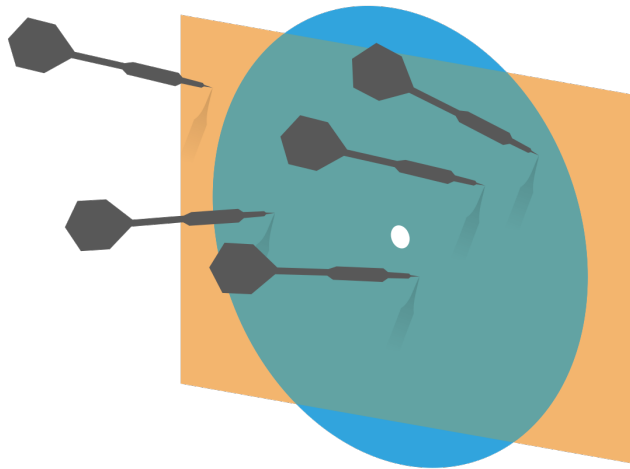


Probability and Statistics for Computer Science



“Correlation is not Causation”
but Correlation is so beautiful!

Credit: wikipedia

PollEv site



PollEv.com/hongyeliu713

Last time



- ✱ Mean
- ✱ Standard deviation
- ✱ Variance
- ✱ Standardizing data
- ✱ Median
- ✱ Percentile

Descriptive stats of 1-D continuous data

Location

Scale

Data set	Mean	Median	Mode	Standard deviation	Variance	Interquartile range
$\{x\}$ or $\{x_i\}$	$\text{mean}(\{x\})$	$\text{median}(\{x\})$	$\text{mode}(\{x\})$	$\text{std}(\{x\})$	$\text{var}(\{x\})$ or $(\text{std}(\{x\}))^2$	interquartile($\{x\}$): 75 percentile - 25 percentile
$\{kx + c\}$	$k * \text{mean}(\{x\}) + c$	$k * \text{median}(\{x\}) + c$	$k * \text{mode}(\{x\}) + c$	$ k * \text{std}(\{x\})$	$k^2 * \text{var}(\{x\})$	$k * \text{interquartile}(\{x\})$ if $k > 0$

Which is more sensitive to outliers?

$$z_i = x_i + y_i$$

$$\text{mean}(\{z_i\}) = \text{mean}(\{x_i\} + \{y_i\})$$

$$= \frac{(x_1 + y_1) + (x_2 + y_2) + \dots + (x_n + y_n)}{n}$$

$$= \frac{n \cdot \text{mean}(\{x_i\}) + n \cdot \text{mean}(\{y_i\})}{n}$$

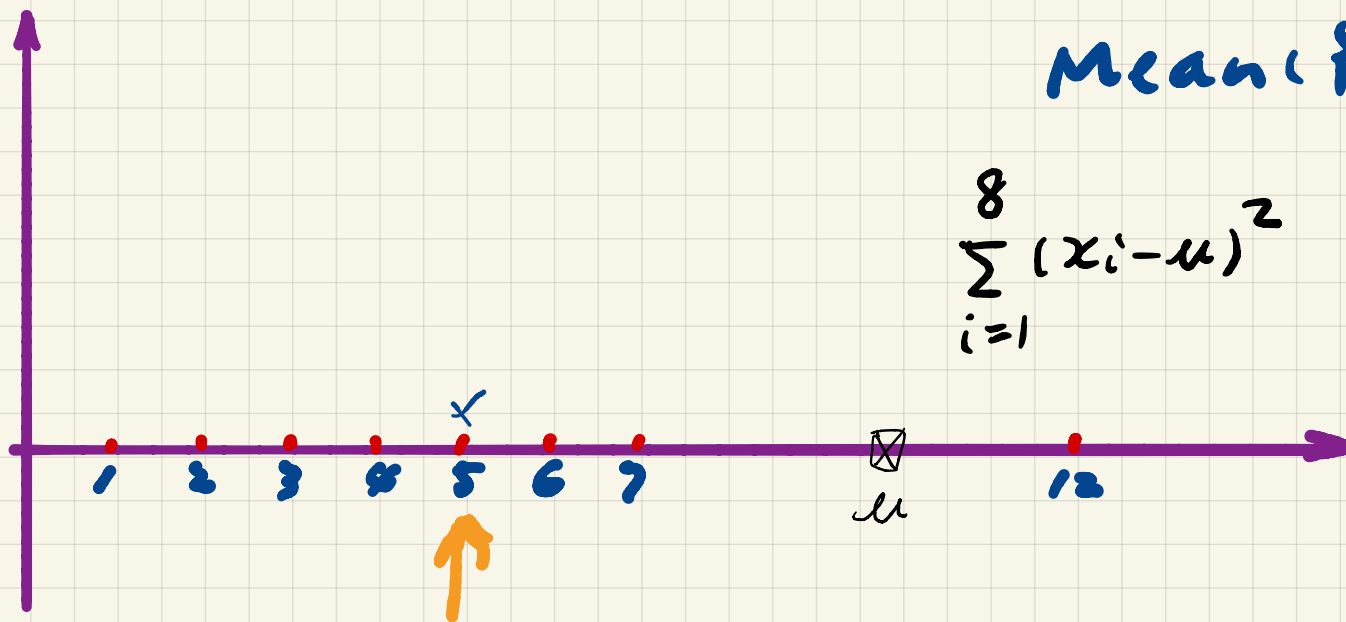
$$= \text{mean}(\{x_i\}) + \text{mean}(\{y_i\})$$

$$\{x_i\} \quad i \in [1, 8]$$

$$\{x_i\} = 1, 2, 3, 4, 5, 6, 7, 12$$

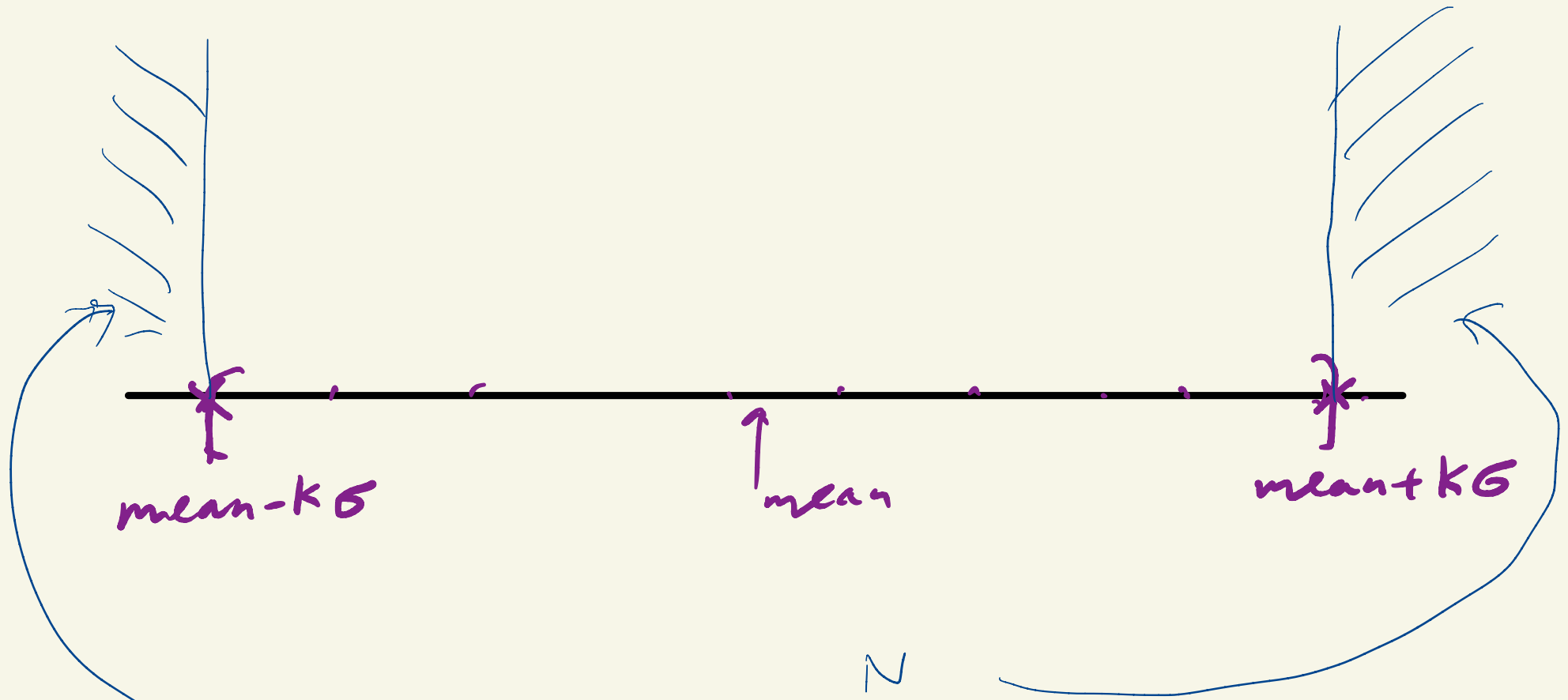
$$\text{Mean}(\{x_i\}) = 5$$

$$\sum_{i=1}^8 (x_i - \mu)^2$$



$\{x_i\} \quad i \in [1, N]$

Chebyshev's
ineq.



$$\# \text{ of pts} \leq \frac{N}{k^2}$$

k is a positive number ≥ 1

σ is the $\text{std}(\{x_i\})$

Q: Estimate the range of data in standard coordinates

✱ The interval $[-5, 5]$ covers $x\%$ data, choose the closest estimate of x .

A. 80%

B. 99

C. 96

D. 96%

$$\hat{x}_i = \frac{x_i - \text{mean}(\{x_i\})}{\text{std}(\{x_i\})}$$

Percentile

✱ k^{th} percentile is the value relative to which $k\%$ of the data items have smaller or equal numbers

✱ Median is roughly the 50^{th} percentile

$\{1, 2, 3, 4, 5, 6, 7, 12\}$

75th percentile = ?

$\neq 0.75$

6

Objectives

- ✱ IQR, Mode
- ✱ Scatter plots for relationships
- ✱ Correlation Coefficient
- ✱ Other Visualization for *relationships*
Heatmap, 3D bar, Time series plots,

Interquartile range

- ✱ $iqr = (75\text{th percentile}) - (25\text{th percentile})$
- ✱ Scaling data scales the interquartile range

$$iqr(\{k \cdot x_i\}) = |k| \cdot iqr(\{x_i\})$$

$k > 0$
 $k \neq 0$

- ✱ Translating data does **NOT** change the interquartile range

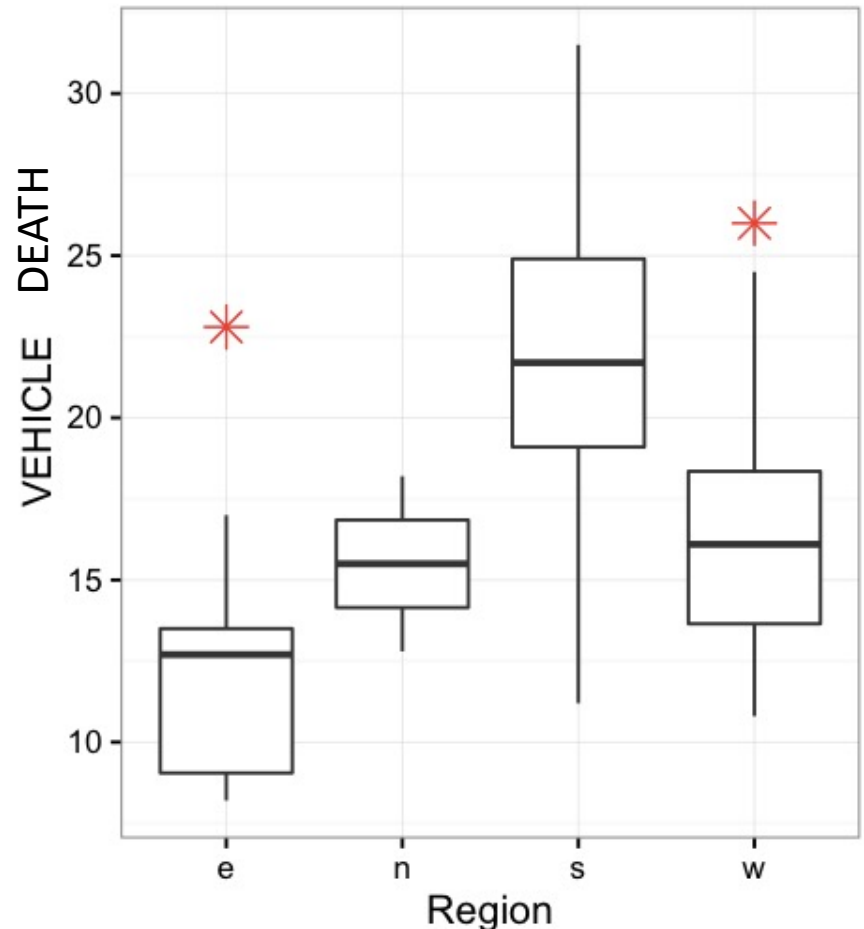
$$iqr(\{x_i + c\}) = iqr(\{x_i\})$$

Box plots

- ✱ Boxplots
 - ✱ Simpler than histogram
 - ✱ Good for outliers
 - ✱ Easier to use for comparison

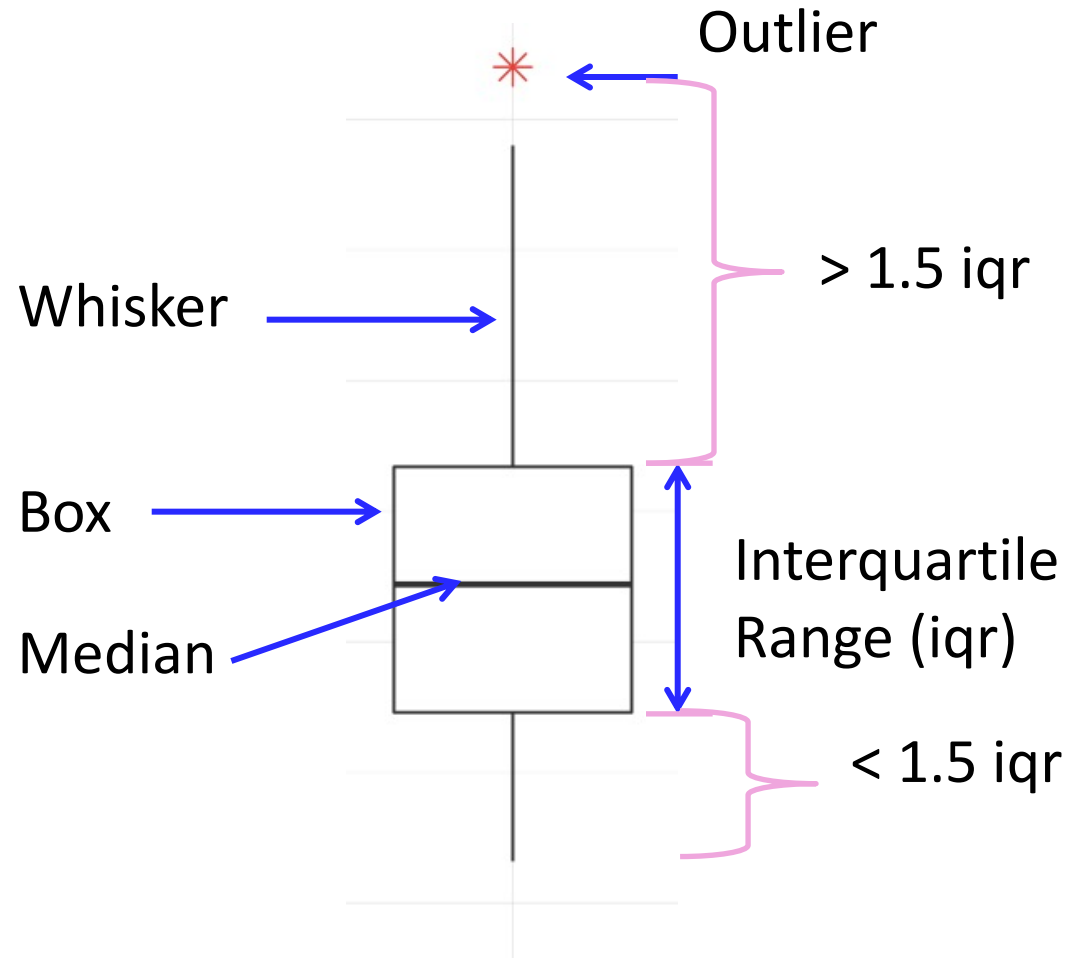
Data from
<https://www2.stetson.edu/~jrasp/data.htm>

Vehicle death by region



Boxplots details, outliers

✱ How to
define
outliers?
(the default)



Sensitivity of summary statistics to outliers

- ✱ mean and standard deviation are very sensitive to outliers
- ✱ median and interquartile range are not sensitive to outliers

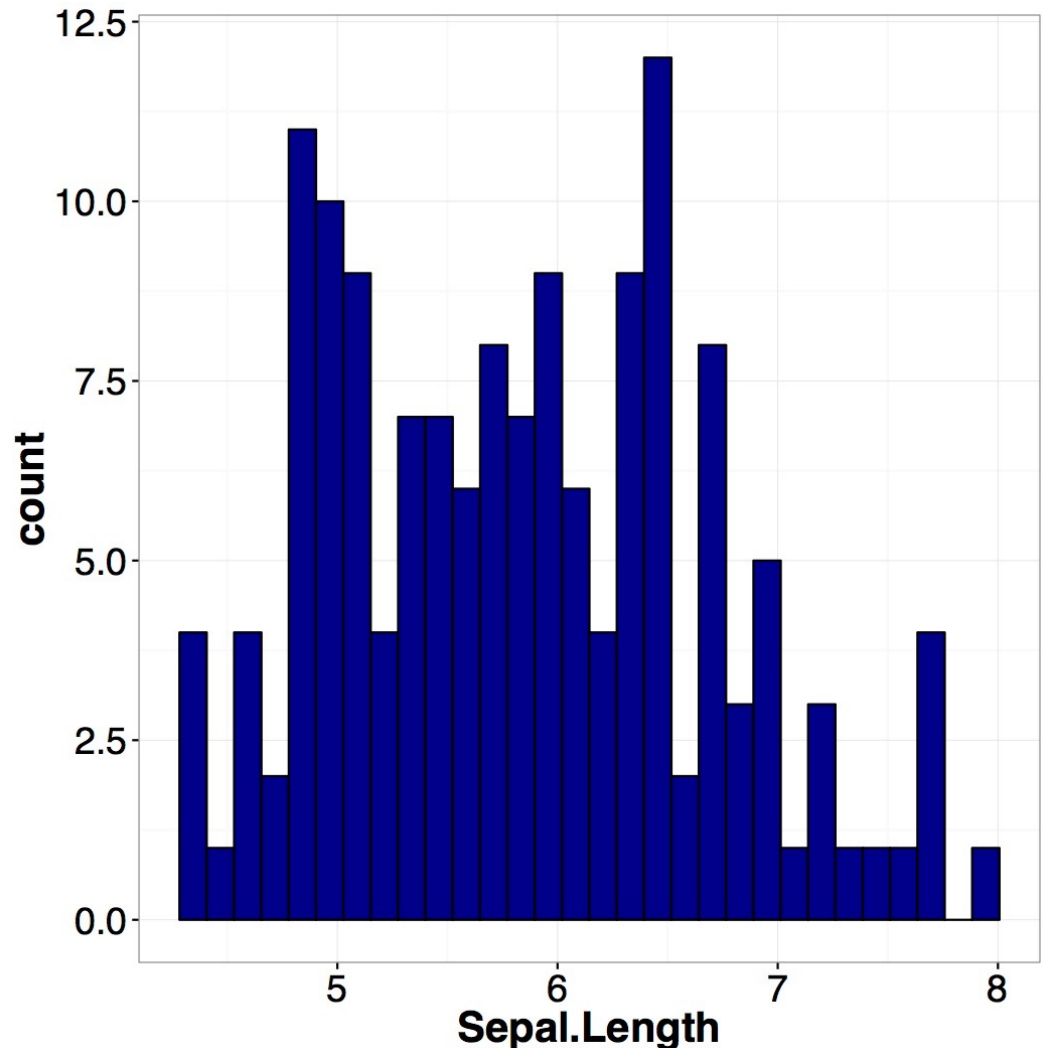
Modes

- ✱ Modes are peaks in a histogram
- ✱ If there are more than 1 mode, we should be curious as to why

Multiple modes

✱ We have seen
the “iris” data
which looks to
have several
peaks

Data: “iris” in R

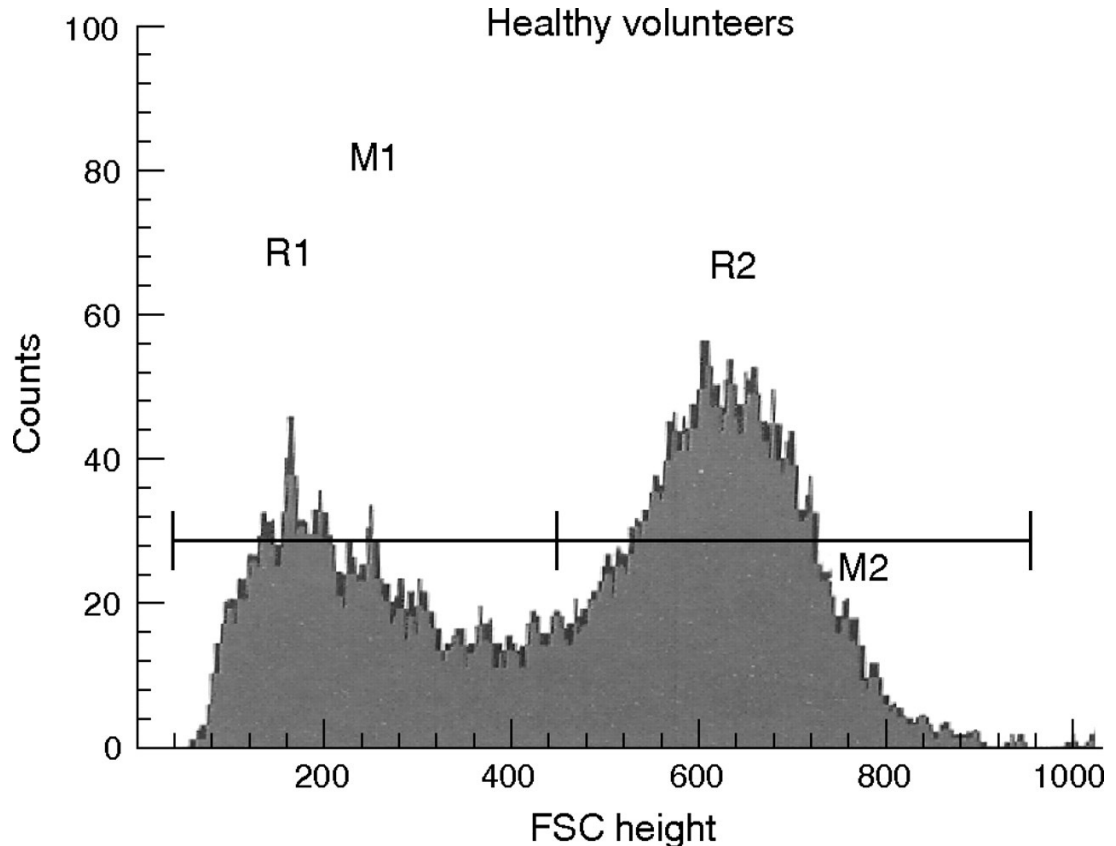


Example Bi-modes distribution

✱ Modes may indicate multiple populations

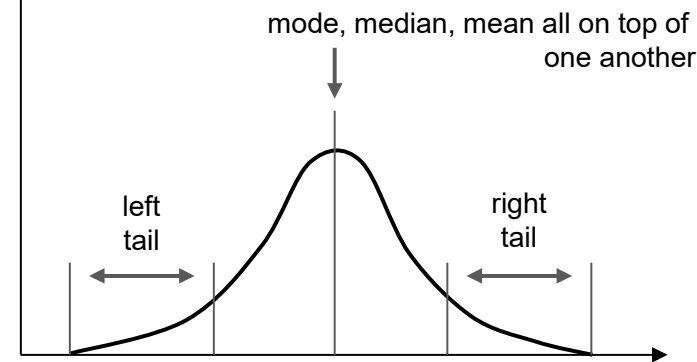
Data: Erythrocyte cells in healthy humans

Piagnerelli, JCP 2007

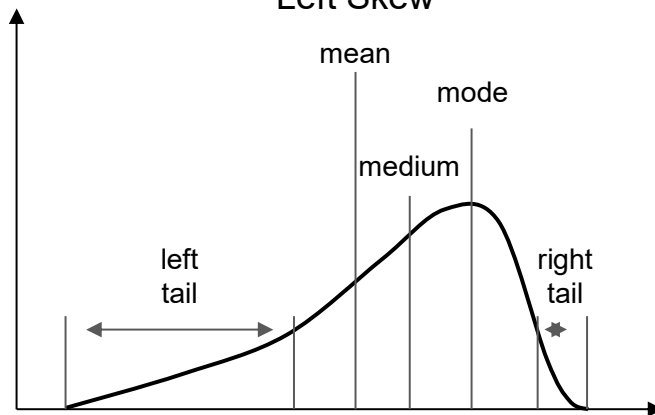


Tails and Skews

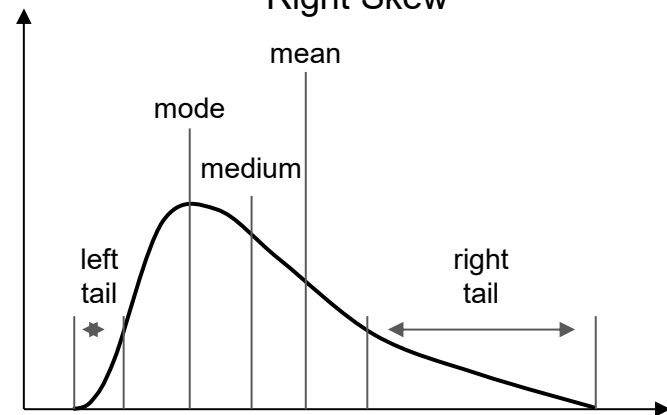
Symmetric Histogram (Smoothed)



Left Skew



Right Skew



Relationship between data features

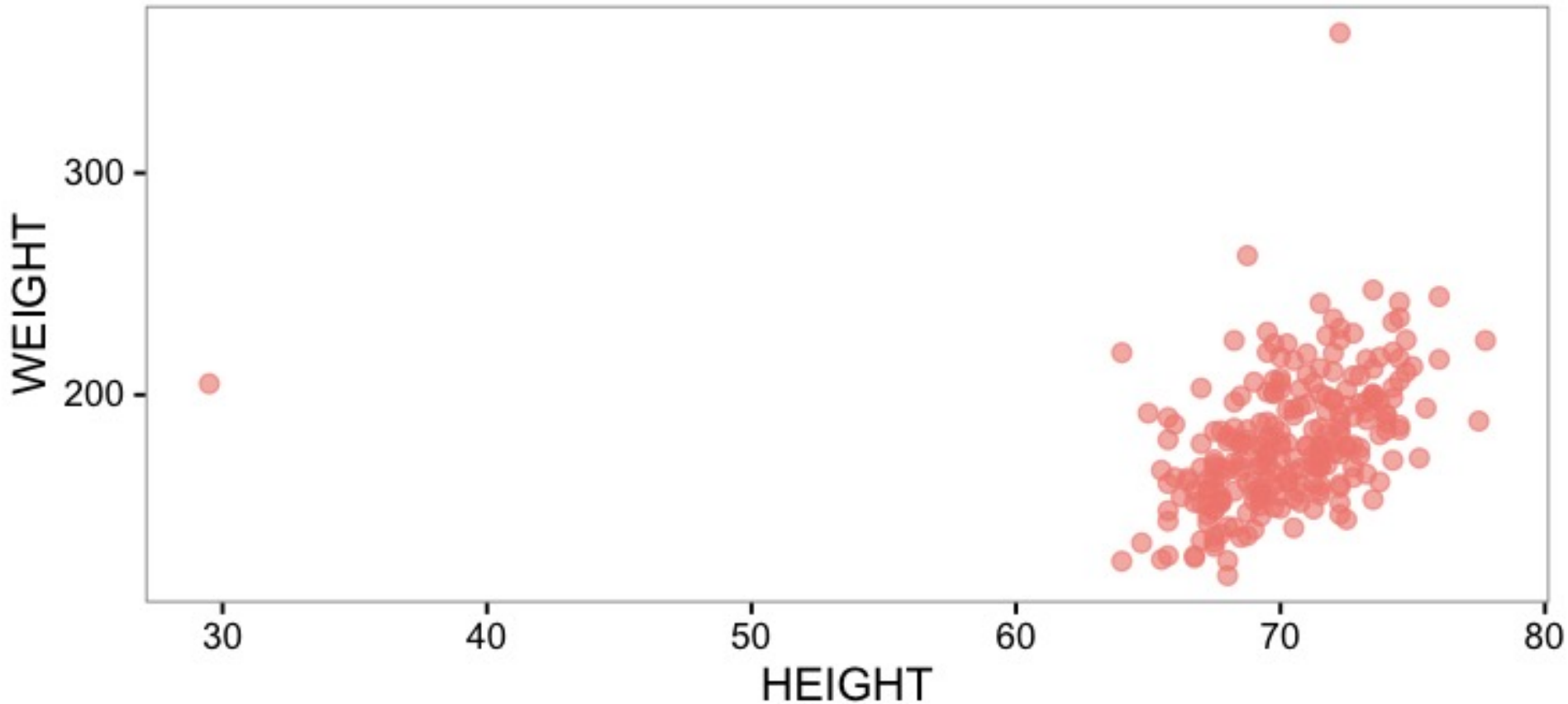
- ✱ Example: Does the weight of people relate to their height?

IDNO	BODYFAT	DENSITY	AGE	WEIGHT	HEIGHT
1	12.6	1.0708	23	154.25	67.75
2	6.9	1.0853	22	173.25	72.25
3	24.6	1.0414	22	154.00	66.25
4	10.9	1.0751	26	184.75	72.25
5	27.8	1.0340	24	184.25	71.25
6	20.6	1.0502	24	210.25	74.75
7	19.0	1.0549	26	181.00	69.75
8	12.8	1.0704	25	176.00	72.50
9	5.1	1.0900	25	191.00	74.00
10	12.0	1.0722	23	198.25	73.50

- ✱ x : HEIGHT, y : WEIGHT

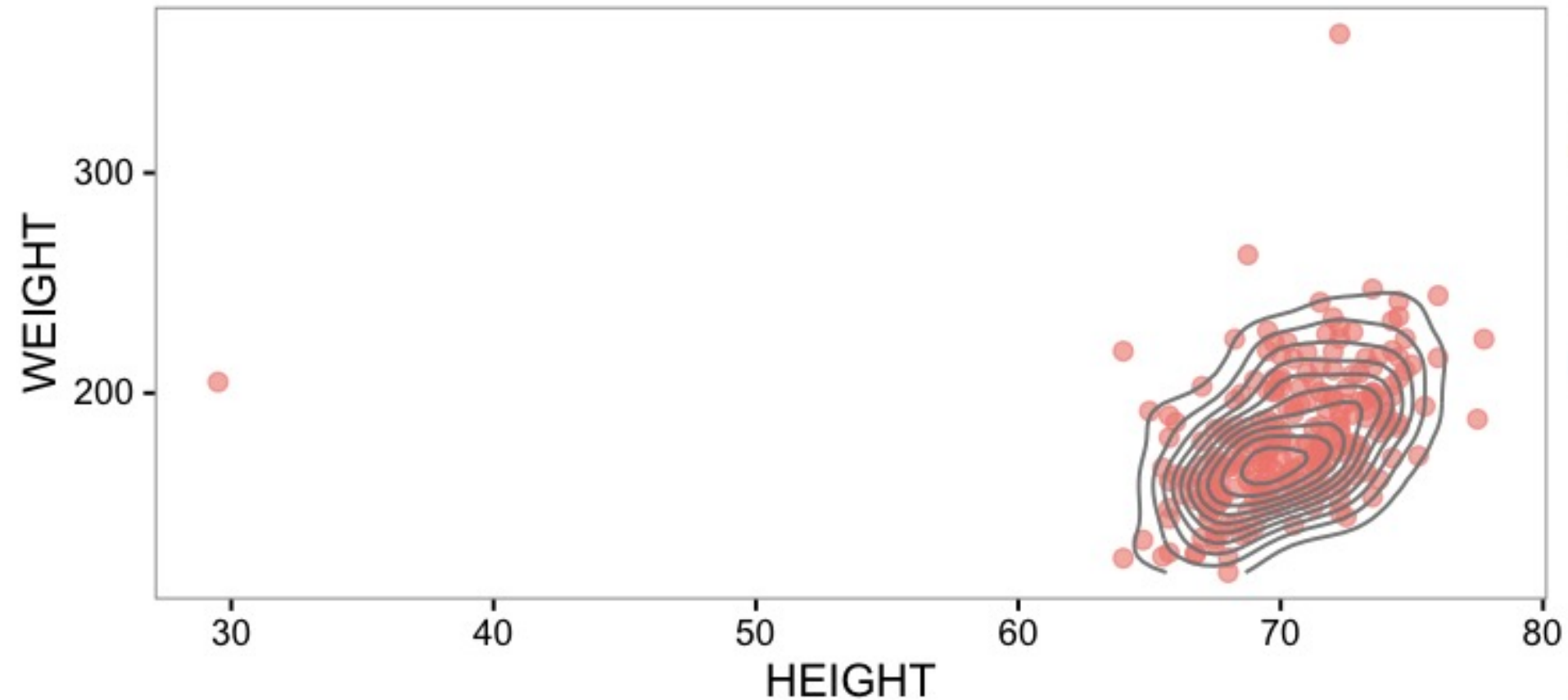
Scatter plot

✱ Body Fat data set



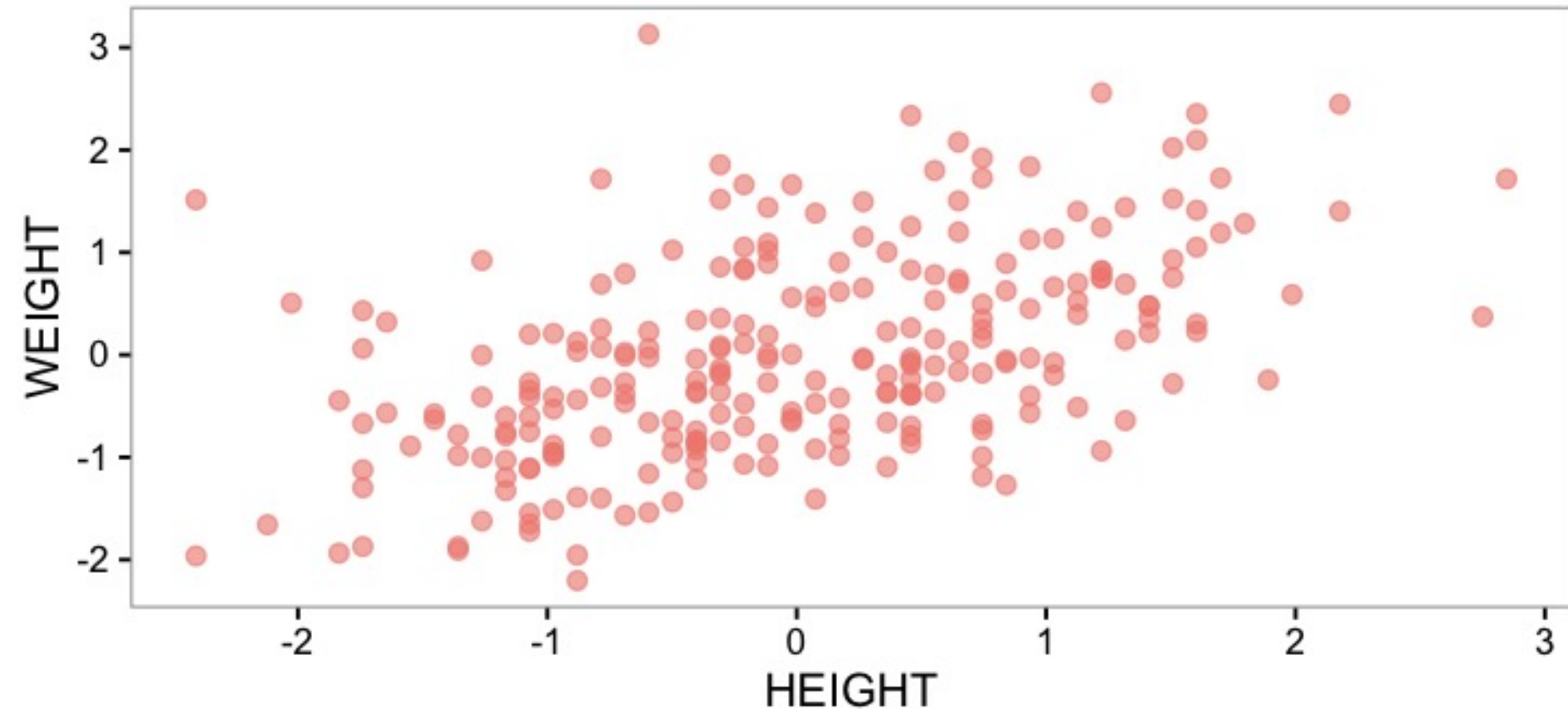
Scatter plot

✳ Scatter plot with density



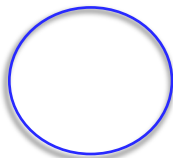
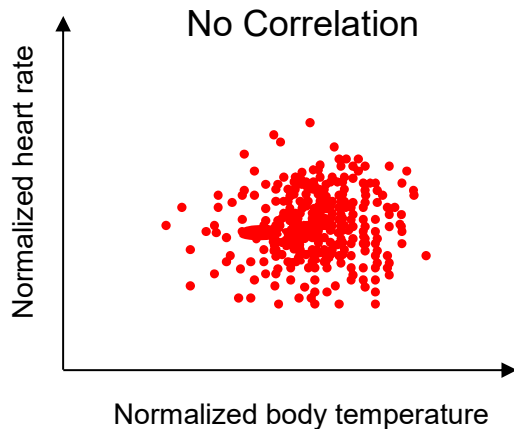
Scatter plot

✱ Removed of outliers & standardized

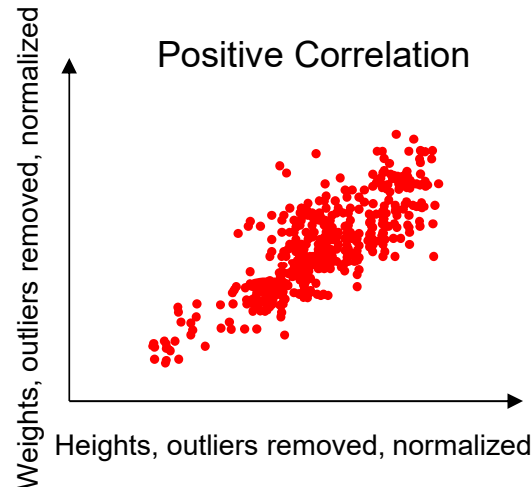


Correlation seen from scatter plots

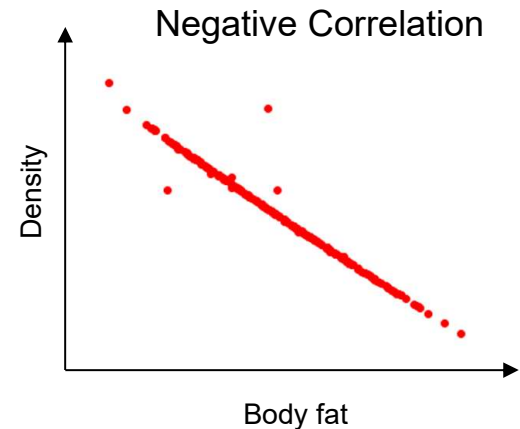
Zero Correlation



Positive correlation



Negative correlation



What kind of Correlation?

- ✱ Line of code in a database and number of bugs
- ✱ Frequency of hand washing and number of germs on your hands
- ✱ GPA and hours spent playing video games
- ✱ earnings and happiness

Correlation Coefficient

✱ Given a data set $\{(x_i, y_i)\}$ consisting of items $(x_1, y_1) \dots (x_N, y_N)$, *2-D data set*

✱ Standardize the coordinates of each feature:

$$\hat{x}_i = \frac{x_i - \text{mean}(\{x_i\})}{\text{std}(\{x_i\})} \quad \hat{y}_i = \frac{y_i - \text{mean}(\{y_i\})}{\text{std}(\{y_i\})}$$

✱ Define the correlation coefficient as:

$$\text{corr}(\{(x_i, y_i)\}) = \frac{1}{N} \sum_{i=1}^N \hat{x}_i \hat{y}_i$$

Correlation Coefficient

$$\hat{x}_i = \frac{x_i - \text{mean}(\{x_i\})}{\text{std}(\{x_i\})}$$

$$\hat{y}_i = \frac{y_i - \text{mean}(\{y_i\})}{\text{std}(\{y_i\})}$$

$$\text{corr}(\{(x_i, y_i)\}) = \frac{1}{N} \sum_{i=1}^N \hat{x}_i \hat{y}_i$$

$$= \text{mean}(\{\hat{x}_i \hat{y}_i\})$$

Q: Correlation Coefficient

✱ Which of the following describe(s) correlation coefficient correctly?

A. It's unitless

B. It's defined in standard coordinates

C. Both A & B

$$\text{corr}(\{(x_i, y_i)\}) = \frac{1}{N} \sum_{i=1}^N \hat{x}_i \hat{y}_i$$

A visualization of correlation coefficient

<https://rpsychologist.com/d3/correlation/>

In a data set $\{(x_i, y_i)\}$ consisting of items $(x_1, y_1) \dots (x_N, y_N)$,

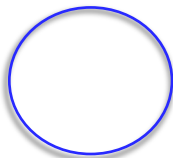
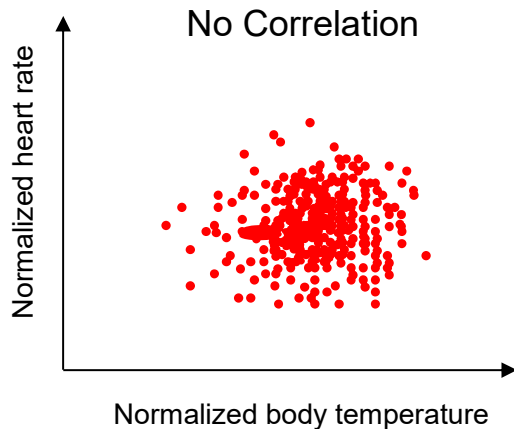
$\text{corr}(\{(x_i, y_i)\}) > 0$ shows positive correlation

$\text{corr}(\{(x_i, y_i)\}) < 0$ shows negative correlation

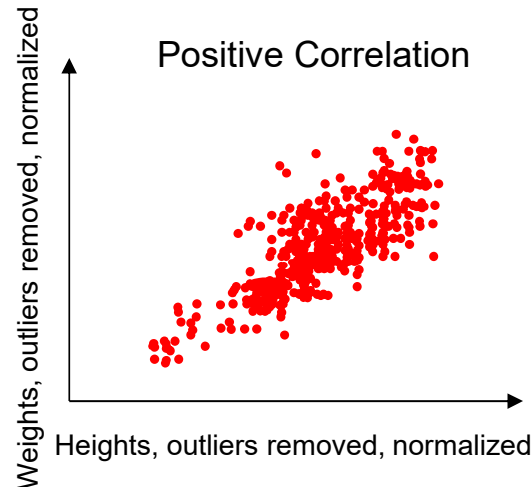
$\text{corr}(\{(x_i, y_i)\}) = 0$ shows no correlation

Correlation seen from scatter plots

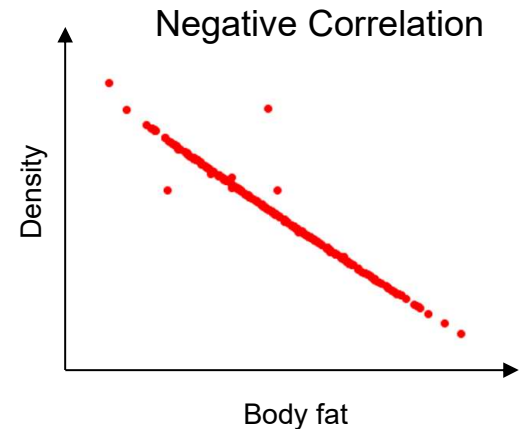
Zero Correlation



Positive correlation



Negative correlation



The Properties of Correlation Coefficient

- ✱ The correlation coefficient is symmetric

$$\text{corr}(\{(x_i, y_i)\}) = \text{corr}(\{(y_i, x_i)\})$$

- ✱ Translating the data does **NOT** change the correlation coefficient

The Properties of Correlation Coefficient

- ✱ Scaling the data may change the sign of the correlation coefficient

$$\begin{aligned} \text{corr}(\{(a x_i + b, c y_i + d)\}) \\ = \text{sign}(a \cdot c) \text{corr}(\{(x_i, y_i)\}) \end{aligned}$$

The Properties of Correlation Coefficient

✱ The correlation coefficient is bounded within $[-1, 1]$

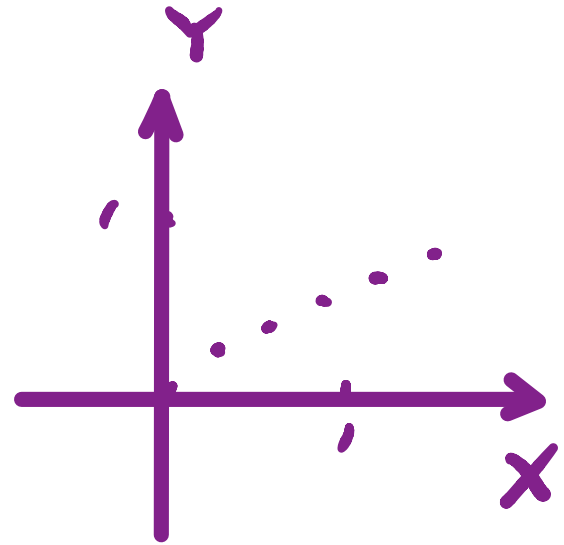
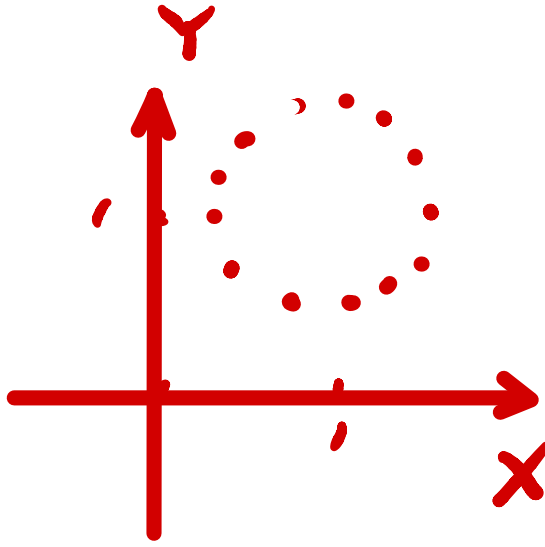
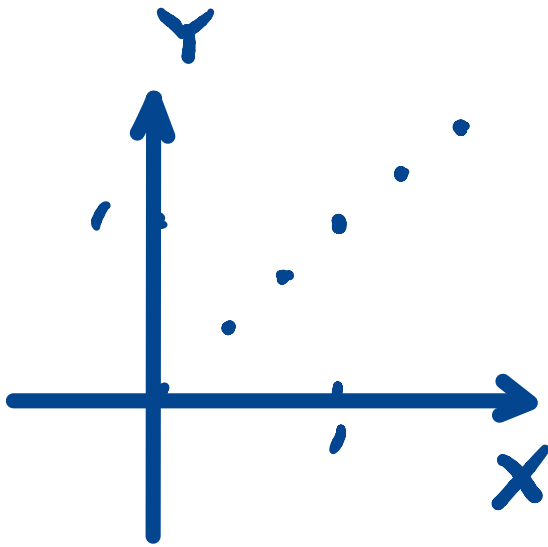
$$\frac{1}{N} \sum_{i=1}^N \hat{x}_i \hat{y}_i$$

$$\{\hat{x}_i\} \quad \{\hat{y}_i\}$$

$corr(\{(x_i, y_i)\}) = 1$ if and only if $\hat{x}_i = \hat{y}_i$

$corr(\{(x_i, y_i)\}) = -1$ if and only if $\hat{x}_i = -\hat{y}_i$

Q. Which of the following has correlation coefficient equal to 1?



- A. Left and right
- B. Left
- C. Middle

Review and
finish it at
home.

$$y = ax; \quad a > 0 \text{ const.}$$

$$\hat{y} = \frac{ax - \text{mean}(\{y\})}{\text{std}(\{y\})}$$

$$= \frac{ax - a \cdot \text{mean}(\{x\})}{a \cdot \text{std}(\{x\})}$$

$$= \hat{x}$$

$$\text{Corr} = \frac{\sum \hat{x}_i \hat{y}_i}{N}$$

$$a > 0$$

what if $y = ax + b$

$$\text{Corr}(x, y) = ?$$

Concept of Correlation Coefficient's bound

- ✱ The correlation coefficient can be written as

$$\text{corr}(\{(x_i, y_i)\}) = \frac{1}{N} \sum_{i=1}^N \hat{x}_i \hat{y}_i$$

$$\text{corr}(\{(x_i, y_i)\}) = \sum_{i=1}^N \frac{\hat{x}_i}{\sqrt{N}} \frac{\hat{y}_i}{\sqrt{N}}$$

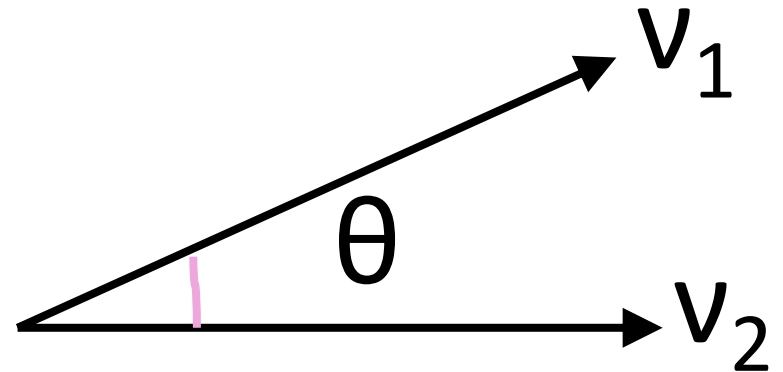
- ✱ It's the inner product of two vectors

$$\left\langle \frac{\hat{x}_1}{\sqrt{N}}, \dots, \frac{\hat{x}_N}{\sqrt{N}} \right\rangle \text{ and } \left\langle \frac{\hat{y}_1}{\sqrt{N}}, \dots, \frac{\hat{y}_N}{\sqrt{N}} \right\rangle$$

Inner product

✱ Inner product's geometric meaning:

$$|\nu_1| |\nu_2| \cos(\theta)$$



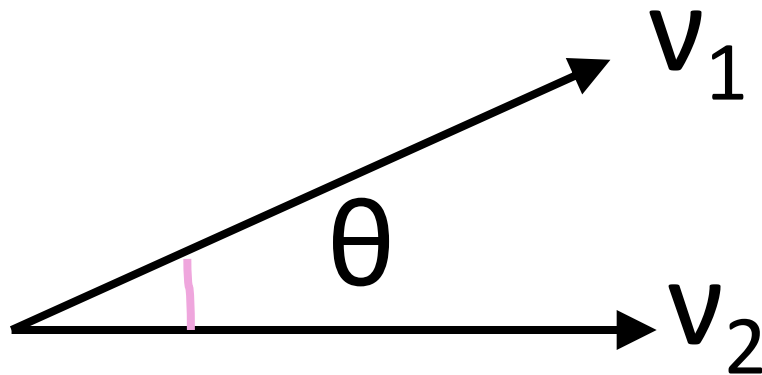
✱ Lengths of both vectors

$$\mathbf{v}_1 = \left\langle \frac{\widehat{x}_1}{\sqrt{N}}, \quad \dots \quad \frac{\widehat{x}_N}{\sqrt{N}} \right\rangle \quad \mathbf{v}_2 = \left\langle \frac{\widehat{y}_1}{\sqrt{N}}, \quad \dots \quad \frac{\widehat{y}_N}{\sqrt{N}} \right\rangle$$

are 1

Bound of correlation coefficient

$$|corr(\{(x_i, y_i)\})| = |\cos(\theta)| \leq 1$$



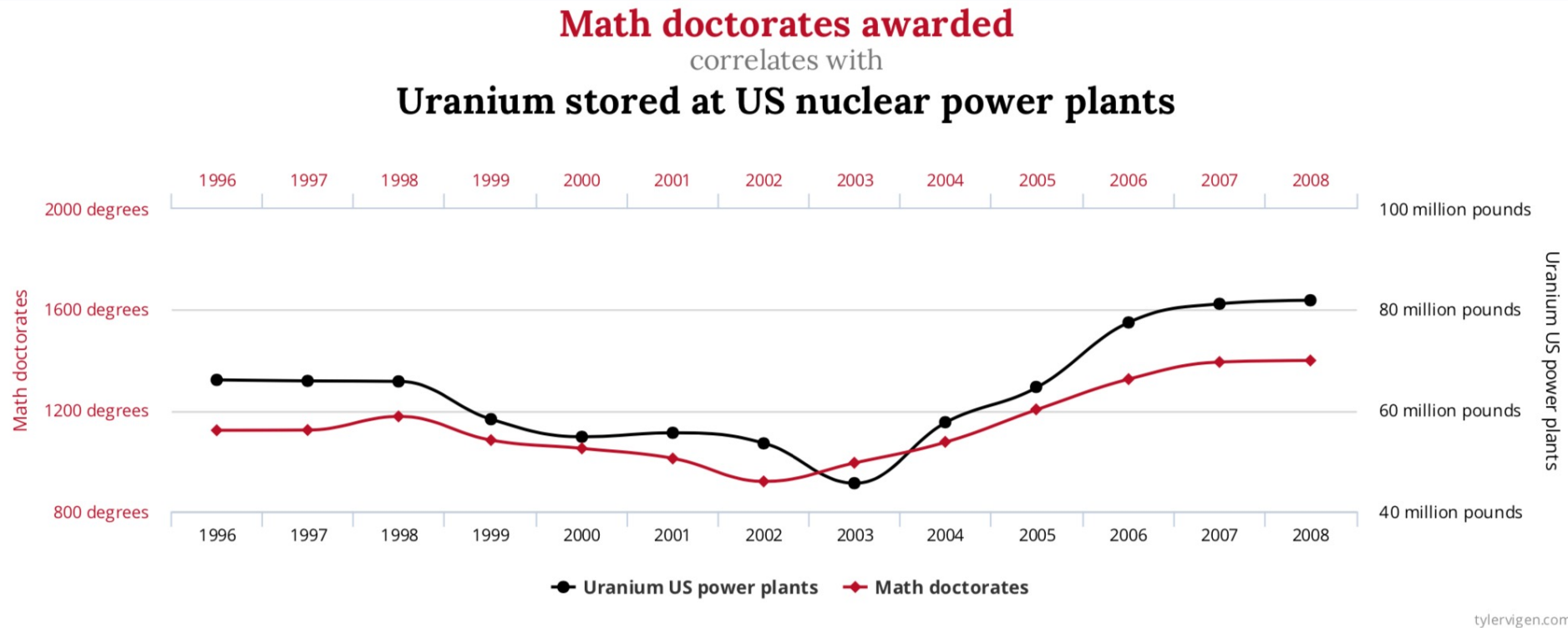
$$v_1 = \left\langle \frac{\widehat{x}_1}{\sqrt{N}}, \quad \dots \quad \frac{\widehat{x}_N}{\sqrt{N}} \right\rangle \quad v_2 = \left\langle \frac{\widehat{y}_1}{\sqrt{N}}, \quad \dots \quad \frac{\widehat{y}_N}{\sqrt{N}} \right\rangle$$

The Properties of Correlation Coefficient

- * Symmetric
- * Translating invariant
- * Scaling only may change sign
- * bounded within $[-1, 1]$

Using correlation to predict

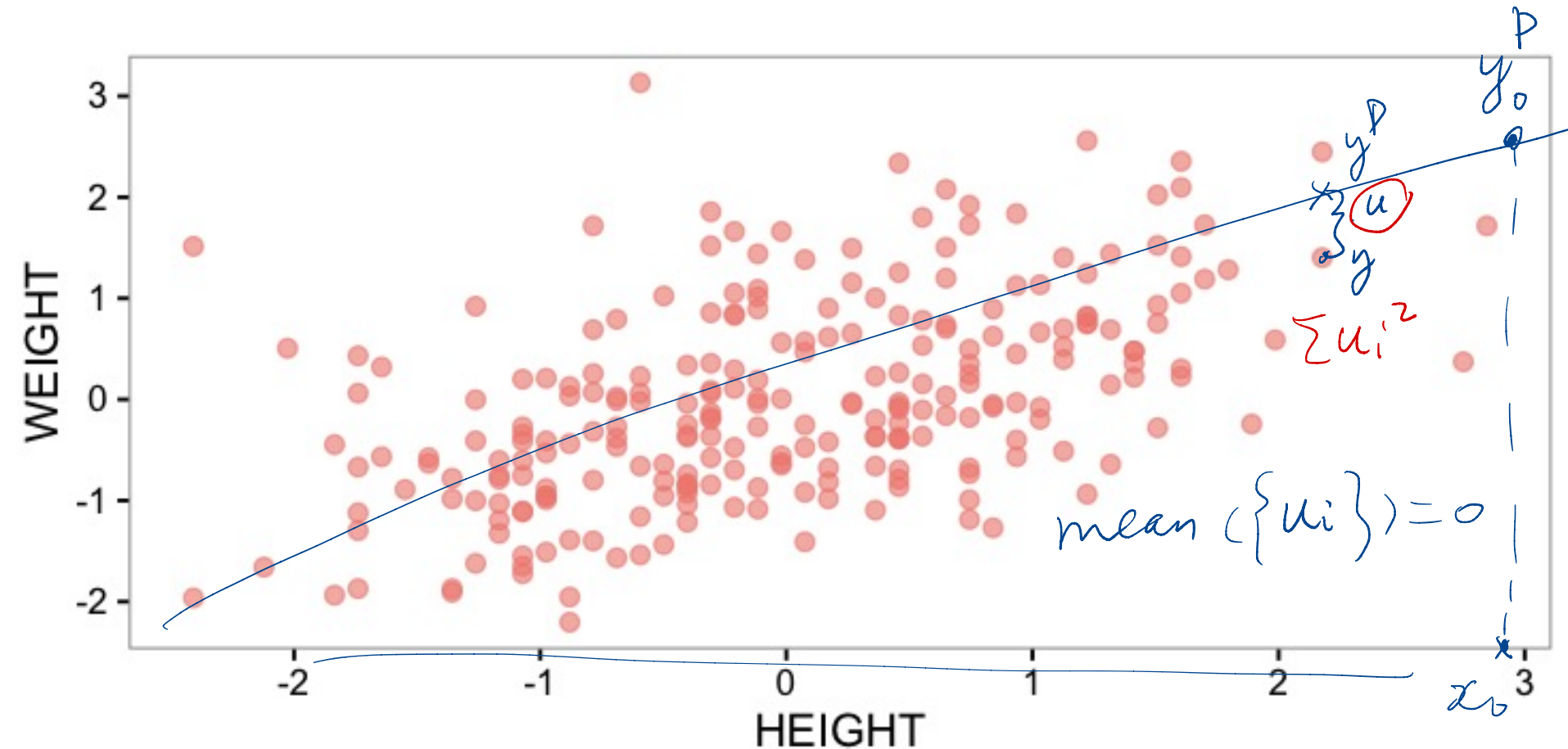
☼ **Caution! Correlation is NOT Causation**



Credit: Tyler Vigen

How do we go about the prediction?

✱ Removed of outliers & standardized



Using correlation to predict

- ✱ Given a correlated data set $\{(x_i, y_i)\}$
we can predict a value y_0^p that goes
with x_0 a value
- ✱ In standard coordinates $\{(\hat{x}_i, \hat{y}_i)\}$
we can predict a value $\hat{y}_0^p$ that goes
with $\hat{x}_0$ a value

Q:

✱ Which coordinates will you use for the predictor using correlation?

A. Standard coordinates

B. Original coordinates

C. Either

Linear predictor and its error

- ✱ We will assume that our predictor is linear

$$\hat{y}^p = a \hat{x} + b$$

- ✱ We denote the prediction at each $\hat{x}_i$ in the data set as $\hat{y}_i^p$

$$\hat{y}_i^p = a \hat{x}_i + b$$

- ✱ The error in the prediction is denoted u_i

$$u_i = \hat{y}_i - \hat{y}_i^p = \hat{y}_i - a \hat{x}_i - b$$

Require the mean of error to be zero



We would try to make the mean of error equal to zero so that it is also centered around 0 as the standardized data:

Require the mean of error to be zero

We would try to make the mean of error equal to zero so that it is also centered around 0 as

the standardized data: $\text{mean}(\{u_i\}) = 0$ ✓

$$\text{mean}(\{u_i\}) = \text{mean}(\{\hat{y} - \hat{y}^p\})$$

$$= \text{mean}(\{\hat{y} - a\hat{x} - b\})$$

Recall:

$$\text{mean}(\{kx + b\})$$

$$= \cancel{\text{mean}(\{\hat{y}\})} - a \cancel{\text{mean}(\{\hat{x}\})} - b$$

$$= k \text{mean}(\{x\}) + b$$

$$= -b = 0$$

$$b = 0$$

$$\text{mean}(\{x_i\} + \{y_i\})$$

$$= \text{mean}(\{x_i\}) + \text{mean}(\{y_i\}) \quad i \in [1, N)$$

Require the variance of error is minimal

minimize

$$\text{var}(\{u_i\})$$

$$\text{var}(\{u_i\}) = \text{mean}(\{u_i - \text{mean}(\{u_i\})\}^2)$$

$$= \text{mean}(\{u_i^2\})$$

$$= \text{mean}(\{\hat{y} - \hat{y}_p\}^2)$$

$$= \text{mean}(\{\hat{y} - a\hat{x}\}^2)$$

$$= \text{mean}(\{\hat{y}^2 - 2a\hat{x}\hat{y} + a^2\hat{x}^2\})$$

$$= \text{mean}(\{\hat{y}^2\}) - 2a\text{mean}(\{\hat{x}\hat{y}\})$$

$$+ a^2\text{mean}(\{\hat{x}^2\})$$

$u_i \rightarrow \text{error}$

Require the variance of error is minimal

minimize

$$\text{var}(\{u_i\})$$

$$\text{var}(\{u_i\}) = \text{mean}(\{u_i - \text{mean}(\{u_i\})\}^2)$$
$$= \text{mean}(\{u_i^2\})$$

$$= \text{mean}(\{(\hat{y} - \hat{y}^P)^2\}) \quad u_i = \hat{y} - \hat{y}^P$$
$$= \text{mean}(\{(\hat{y} - a\hat{x})^2\}) \quad = \hat{y} - a\hat{x} \quad \because b=0$$

$$= \text{mean}(\{\hat{y}^2 - 2a\hat{x}\hat{y} + a^2\hat{x}^2\})$$

$$= \text{mean}(\{\hat{y}^2\}) - 2a \text{mean}(\{\hat{x}\hat{y}\}) + a^2 \text{mean}(\{\hat{x}^2\})$$

$$\text{mean}(\{\hat{y}^2\})$$

$$= \text{mean}(\{(\hat{y} - 0)^2\})$$

$$= \text{mean}(\{(\hat{y} - \text{mean}(\{\hat{y}\})\}^2\})$$

$$= \text{var}(\{\hat{y}\}) = 1$$

$$\text{mean}(\{\hat{x}^2\}) = 1$$

corr $\rightarrow r$

Require the variance of error is minimal

$$\text{var}(\{u\}) = \text{mean}(\{\hat{y}^2\}) - 2a \text{mean}(\{\hat{x} \hat{y}\}) + a^2 \text{mean}(\{\hat{x}^2\})$$

$$= 1 - 2a \text{mean}(\{\hat{x} \hat{y}\}) + a^2$$

$$= 1 - 2a \text{corr}(\{x, y\}) + a^2$$

$$\text{let } r = \text{corr}(\{x, y\})$$

$$= 1 - 2ar + a^2$$

$$\frac{d(\text{var}(\{u\}))}{da} = -2r + 2a = 0$$

$a = r$

$\text{var}(\{u\})$

$\text{argmin}_a (\downarrow)$

Require the variance of error is minimal

recall : $\hat{y}^p = a\hat{x} + b$

Now we found : $b = 0$ $a = r = \text{corr}$

So : $\hat{y}^p = r \cdot \hat{x}$

Here is the linear predictor!

$$\hat{y}^p = r \hat{x}$$



Correlation coefficient

Prediction Formula

✱ In standard coordinates

$$\hat{y}_0^p = r \hat{x}_0 \quad \text{where} \quad r = \text{corr}(\{(x_i, y_i)\})$$

✱ In original coordinates

$$\frac{y_0^p - \text{mean}(\{y_i\})}{\text{std}(\{y_i\})} = r \frac{x_0 - \text{mean}(\{x_i\})}{\text{std}(\{x_i\})}$$

Root-mean-square (RMS) prediction error



Given $var(\{u_i\}) = 1 - 2ar + a^2$
& $a = r$

$$var(\{u_i\}) = 1 - r^2$$



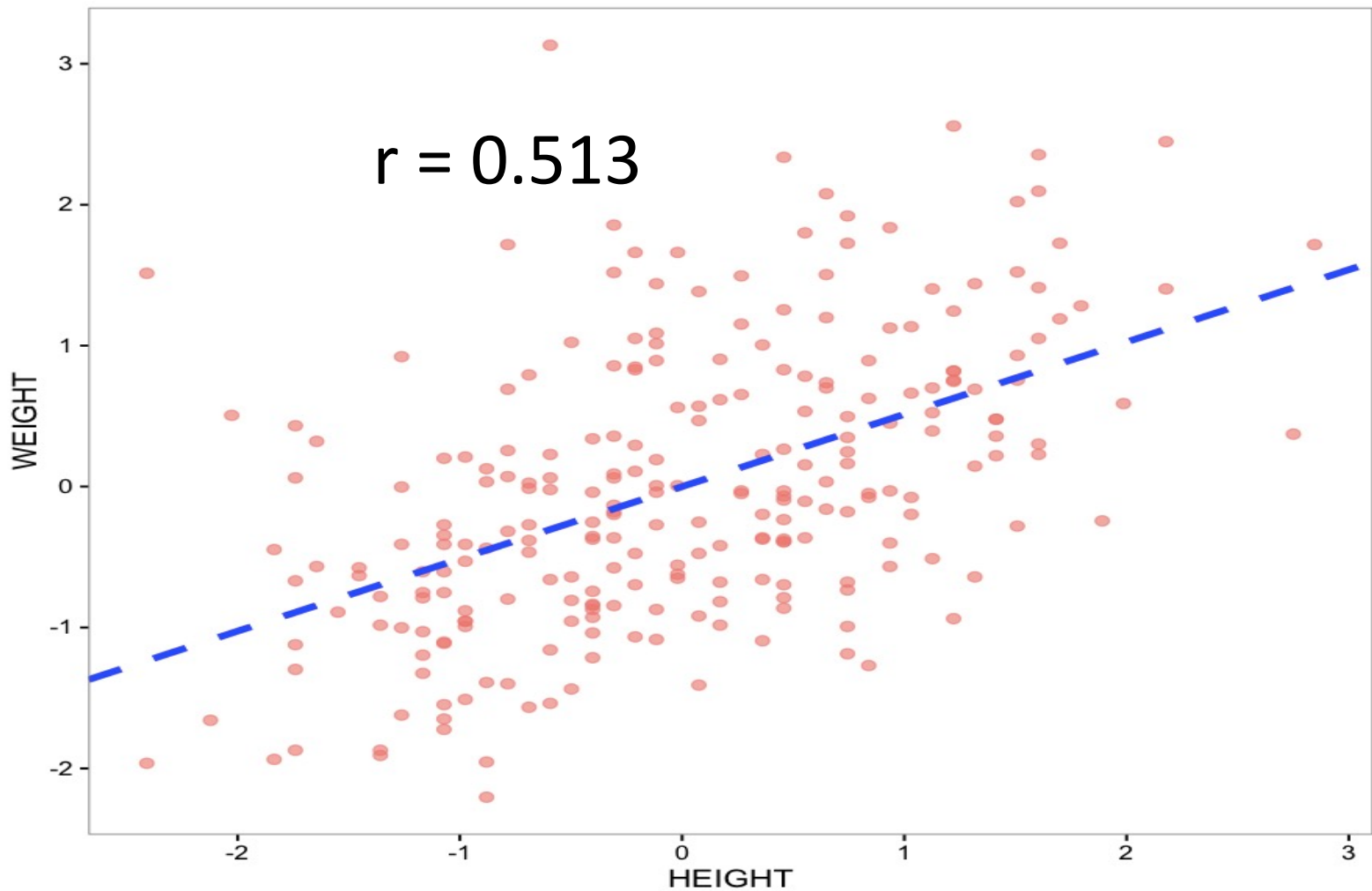
$$\begin{aligned} RMS \text{ error} &= \sqrt{mean(\{u_i^2\})} \\ &= \sqrt{var(\{u_i\})} \\ &= \sqrt{1 - r^2} \end{aligned}$$

See the error through simulation

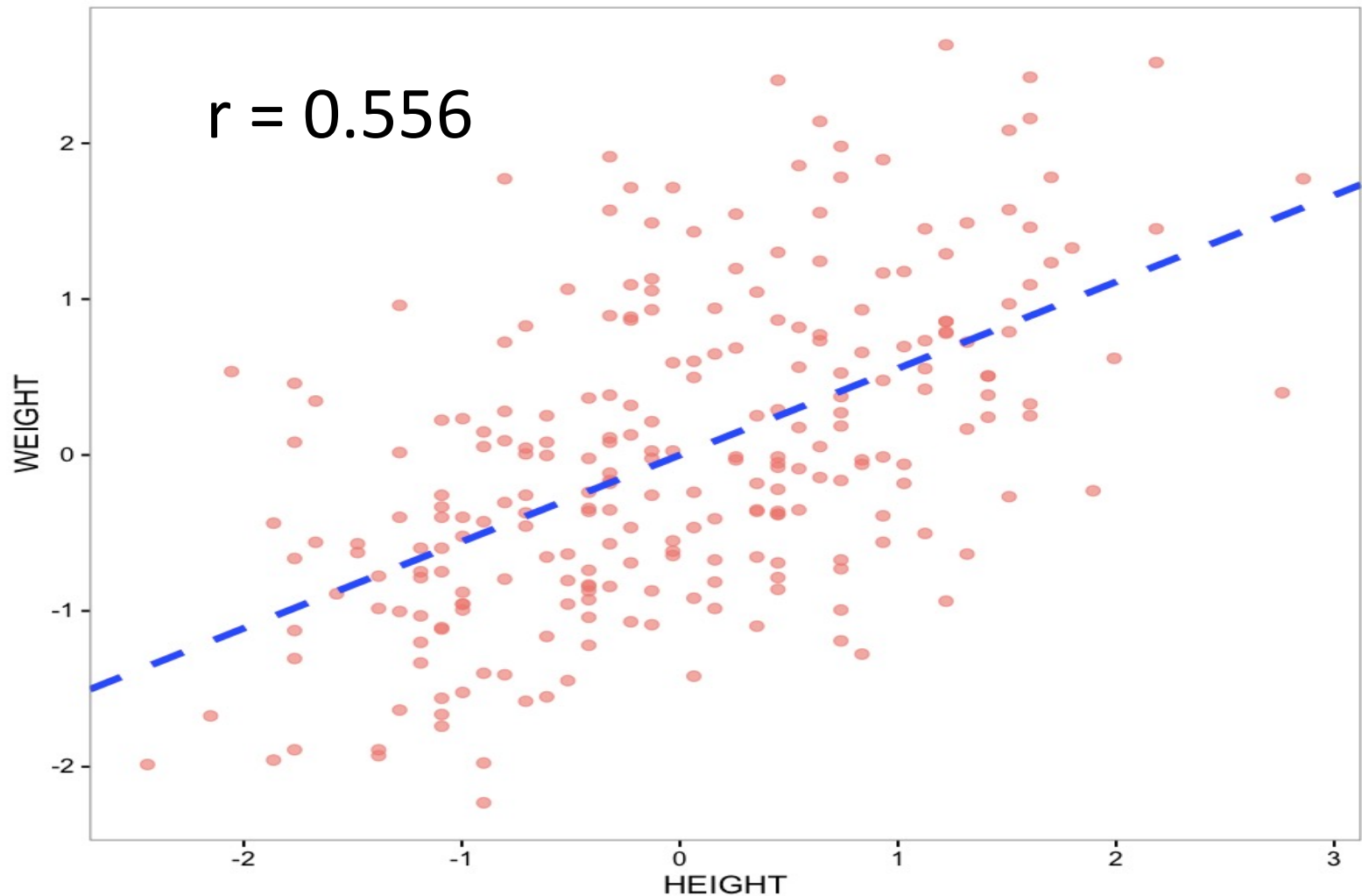


<https://rpsychologist.com/d3/correlation/>

Example: Body Fat data

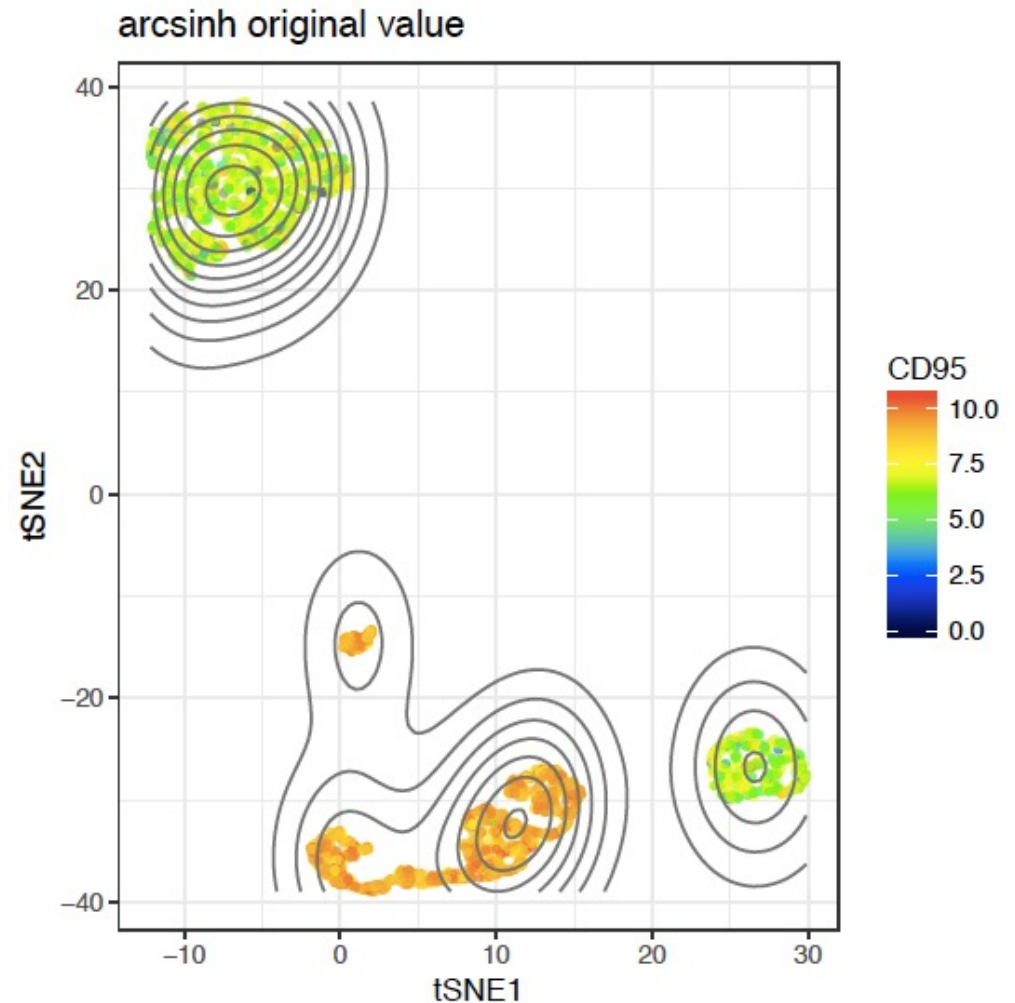


Example: remove 2 more outliers

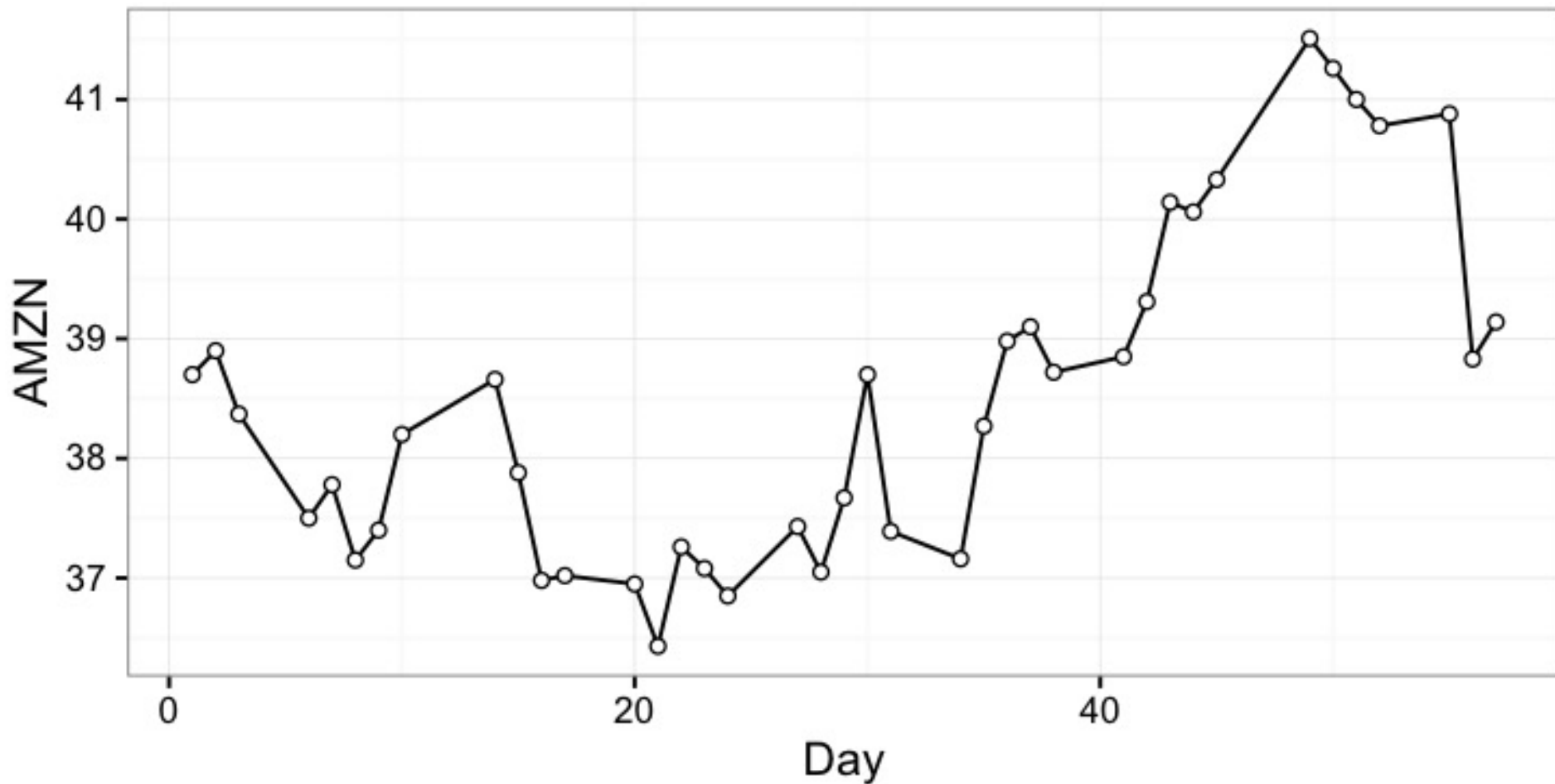


Scatter plot

✳ Coupled with heatmap to show a 3rd feature



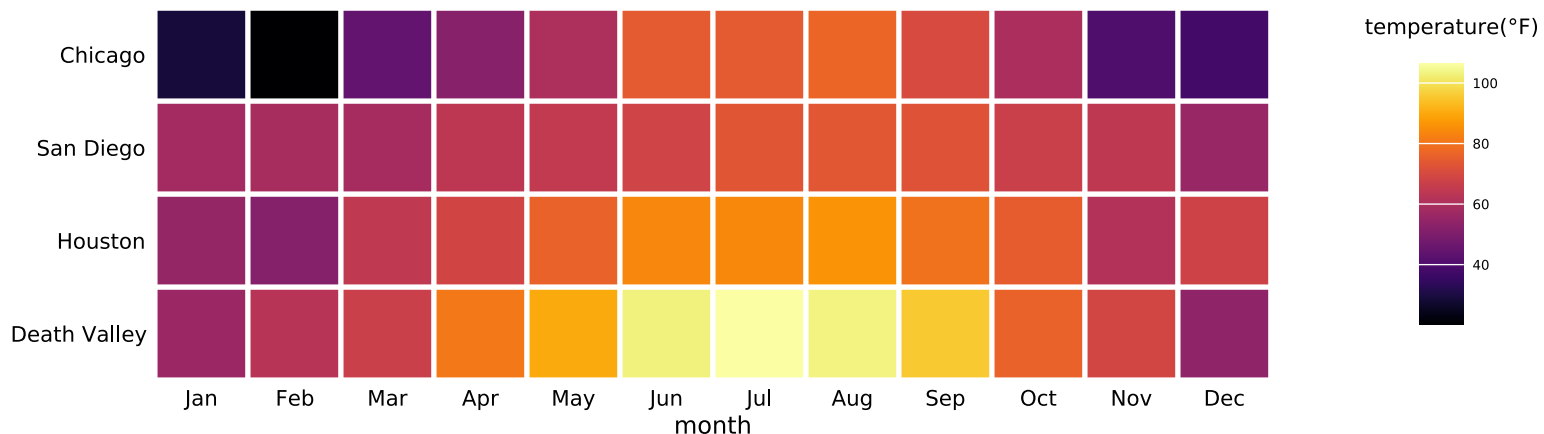
Time Series Plot: Stock of Amazon



Heatmap

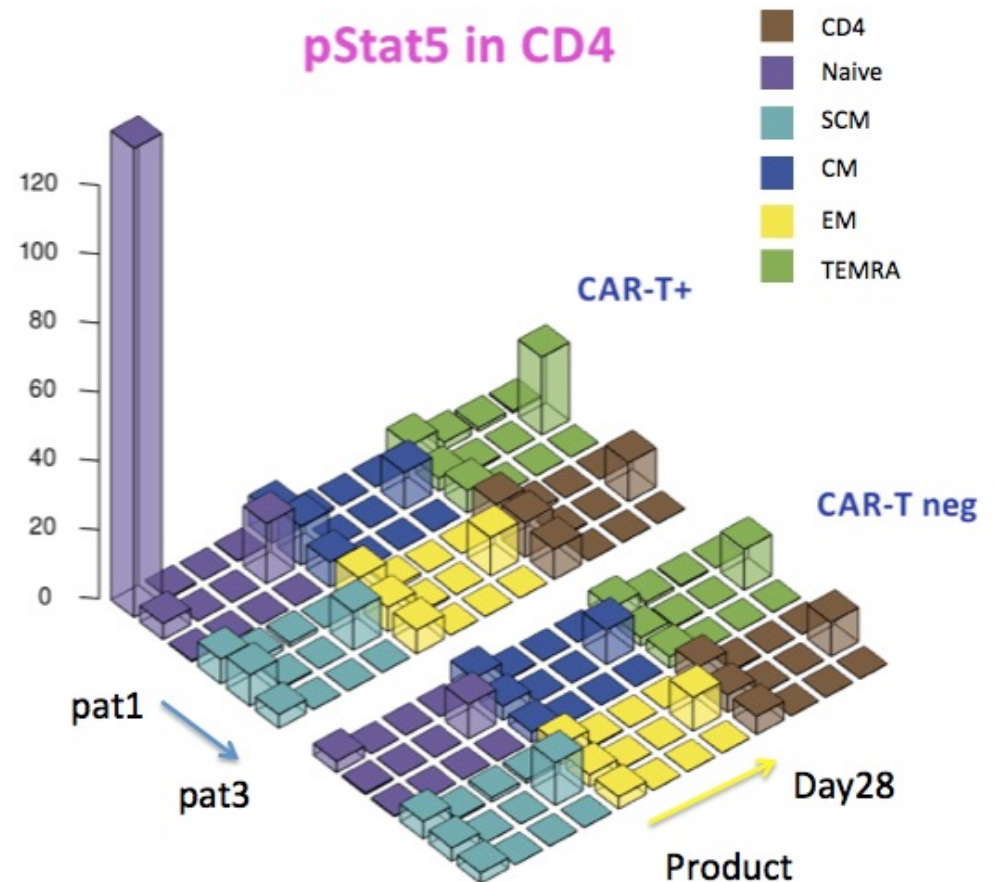
✱ Display matrix of data via gradient of color(s)

Summarization of 4 locations' annual mean temperature by month



3D bar chart

✱ Transparent
3D bar chart
is good for
small # of
samples
across
categories



Assignments

- ✱ Quiz1 open at 4:30pm today on PL
- ✱ Finish reading Chapter 2 of the textbook
- ✱ Work on the Week 2 module on Canvas
- ✱ Next time: Probability a first look

Additional References

- ✱ Charles M. Grinstead and J. Laurie Snell
"Introduction to Probability"
- ✱ Morris H. Degroot and Mark J. Schervish
"Probability and Statistics"

See you next time

*See
You!*

